

Universal Plus Technical Report



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1. Academic Paper Discussion

The literature review supported the project's methodological choices by synthesising six key academic and industry papers addressing three central challenges: class imbalance, high dimensionality, and model selection for credit-related prediction problems.

To address class imbalance, the review contrasted findings from Suguna et al. (2025) and Elreedy and Atiya (2019). Suguna et al. (2025) demonstrated that Synthetic Minority Over-sampling Technique (SMOTE) combined with ensemble methods improves recall for rare events such as churn. However, Elreedy and Atiya (2019) provided mathematical evidence that SMOTE deteriorates in high-dimensional feature spaces due to the “curse of dimensionality” and produces synthetic samples that do not represent the true minority distribution. Given this project's 200 anonymised variables, this limitation was decisive; SMOTE was intentionally avoided. Instead, random undersampling and cost-sensitive learning were prioritised to maintain distributional integrity.

Regarding model selection, Nittner (2009) highlighted the regulatory relevance and interpretability of linear models in credit risk, which justified Logistic Regression as the project's transparent baseline. Conversely, Pisner and Schnyer (2020) noted the theoretical strengths of Support Vector Machines (SVMs) in high-dimensional contexts but also acknowledged their high computational cost and limited interpretability—making them less practical for deployment.

Finally, findings from Prajwala (2015) and Dwidarma et al. (2021) demonstrated the superiority of ensemble tree methods. Prajwala (2015) showed that Random Forests outperform single decision trees by mitigating variance through bagging. Dwidarma et al. (2021) further revealed that eXtreme Gradient Boosting (XGBoost) outperforms Logistic Regression for debtor prediction tasks, especially in detecting minority-class outcomes.

The literature ultimately supported selecting an undersampled Light Gradient Boosting Model (LightGBM) model with gain-based feature pruning as the final champion architecture.

2. Introduction

Universal Plus aims to proactively identify customers who are likely to apply for business credit in the near future. Accurately anticipating these applications would allow relationship managers to tailor outreach, improve customer experience, and strengthen engagement. With increasing competition in the financial sector, capability in predictive customer behaviour modelling has become a strategic differentiator. A survey conducted by the Bank of England in 2024 found that 75% of UK financial institutions already use artificial intelligence, with a further 10% planning adoption within three years.

Predictive analytics, particularly machine learning, has become central to decision-making in financial services. The literature highlights applications for strategic purposes ranging from customer segmentation and retention to creditworthiness prediction (Broby, 2022). Prominent machine learning models such as gradient boosting architectures have demonstrated considerable accuracy in risk forecasting with AUC scores reaching up to 0.94 in some credit decisioning contexts (Byrne, 2025).

This project aims to develop a data-driven predictive pipeline using 100,000 customer records and 200 anonymised numerical features to predict future credit applications. The CRISP-DM framework was adopted to structure the work. Initial steps involved assessing data quality, completeness, and imbalance. Subsequent work focused on preprocessing, feature engineering, and comparing a suite of machine learning models including Logistic Regression, Decision Trees, Random Forests, XGBoost, and LightGBM.

The project concludes by evaluating final model performance, operational implications, and opportunities for future enhancement. The final model is an undersampled LightGBM with gain-based feature pruning that demonstrates strong predictive power and operational efficiency, positioning it as a practical tool for Universal Plus to strengthen predictive outreach strategies.

3. Methodology

The modelling methodology followed CRISP-DM and was designed to systematically address the challenges posed by a highly imbalanced, high-dimensional dataset. The dataset consists of 200 anonymised behavioural and numerical features across 100,000 customer records, whereas only around 10% belongs to the positive class.

3.1 Data Understanding and Preliminary Exploration

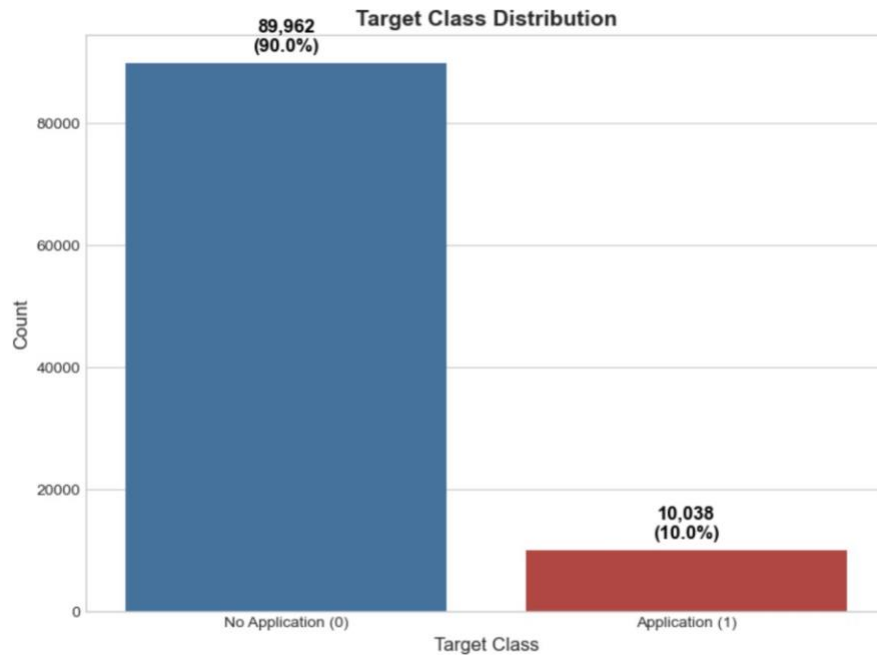


Figure 1. Distribution of the target variable showing substantial class imbalance

This imbalance immediately justified the need for specialised handling during modelling, as standard algorithms would otherwise predict overwhelmingly negative outcomes.

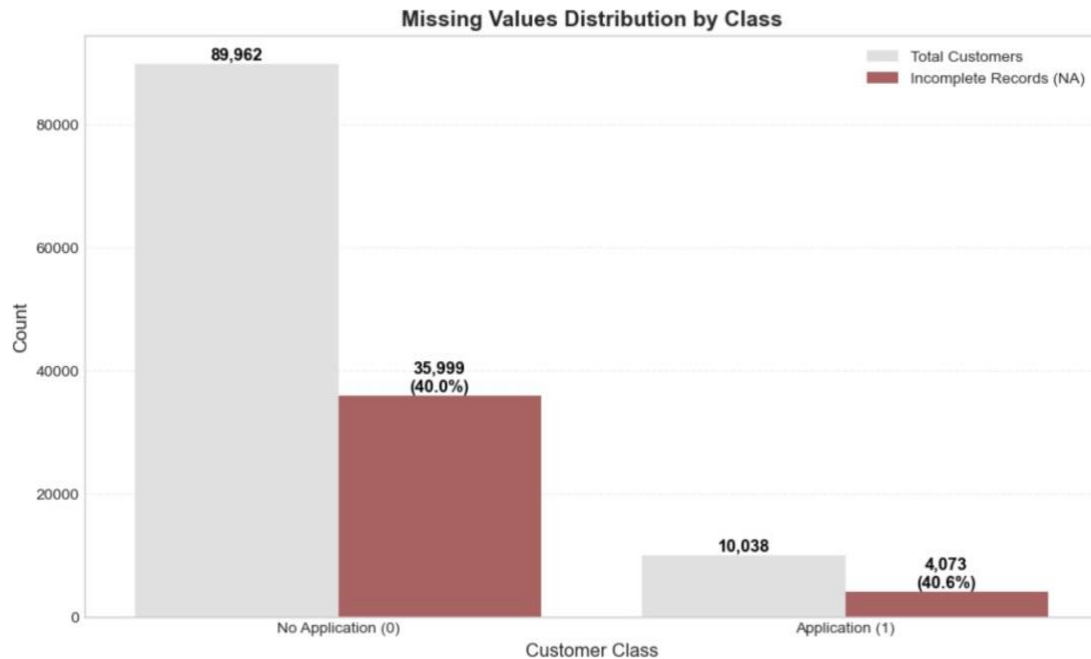


Figure 2. Distribution of missing values across the dataset, showing non-uniform missingness

3.2 Data Preparation

Key preprocessing steps included:

- Removing Non-Predictive Identifiers

ID_code was dropped because it provided no behavioural information for the predictive model.

- Outlier Detection and Removal

Z-score filtering ($|z| > 4$) identified and removed 11 extreme rows, preventing distortions during model training.

- Train-Test Split (80/20, Stratified):

A stratified split was implemented to ensure the minority class remained proportionally represented in both sets.

- Mean Imputation

A SimpleImputer was fitted on training data only and applied to the test set to avoid information leakage.

SMOTE was intentionally avoided in favour of random undersampling, as research by Elreedy & Atiya (2019) provides mathematical evidence that synthetic oversampling deteriorates in high-dimensional feature spaces by generating samples that do not represent the true minority distribution.

Our modelling pipeline employed two distinct strategies to address the 90:10 class

imbalance, tailored to the specific needs of each model stage:

1. Baseline & Challenger (Logistic Regression / XGBoost): Trained on the full imbalanced dataset using `scale_pos_weight` to mathematically penalize false negatives (9×), simulating balance without modifying the data structure.
2. Champion Model (LightGBM): Transitioned to Random Undersampling to create a physical 50:50 distribution. Consequently, `scale_pos_weight` was removed to prevent a "double correction" bias toward false positives, ensuring the model learned from a naturally balanced signal.

3.3 Model Tested and Learnings

Note on Class Definitions & Selection Criteria: In this analysis, Class 0 represents customers who do not apply for a loan (Majority), while Class 1 represents those who do (Minority). Since all models naturally perform well on the dominant Class 0, overall accuracy is a poor metric. Therefore, model selection was driven strictly by performance on Class 1.

Hypothesis Model 1: Baseline Modelling - Logistic Regression + Scale Weight

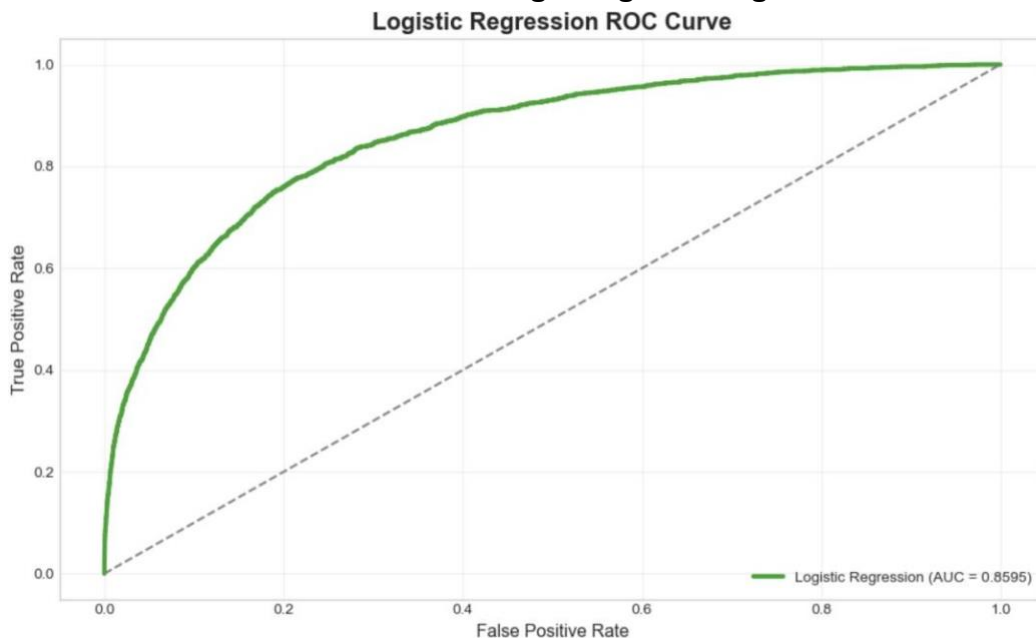


Figure 3. ROC curve for Logistic Regression

Logistic Regression was selected as the regulatory-friendly baseline to establish a linear performance benchmark. To address the severe class imbalance without altering the data

distribution, we implemented Cost-Sensitive Learning using the 'scale_pos_weight' parameter. This mathematically adjusted the loss function to penalise false negatives 9× more heavily than false positives, effectively simulating a balanced dataset during gradient descent.

Resulting AUC: 0.8595

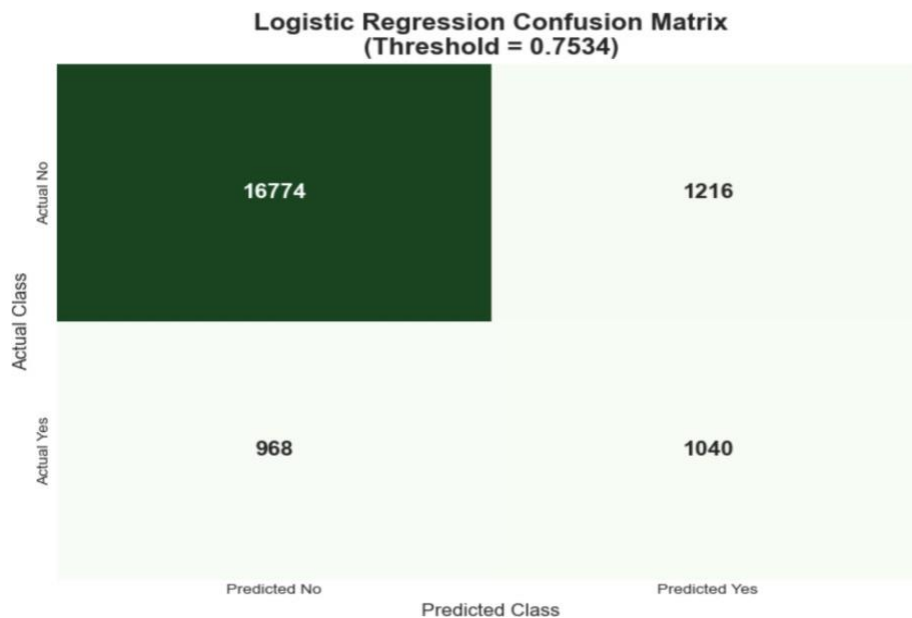


Figure 4. Confusion matrix at the optimised F1 threshold

Hypothesis Model 2: Feature Engineering: Statistical + Frequency Encoding + Scale Weight

To enhance predictive richness despite the anonymity of the variables, the engineering pipeline was expanded to include row-level statistical features (mean, skewness, variance) and frequency-encoded columns. These engineered features were trained using an eXtreme Gradient Boosting (XGBoost) architecture. Similar to the baseline, this model utilised 'scale_pos_weight' to handle class imbalance while leveraging a level-wise tree growth strategy to capture non-linear feature interactions.

Resulting AUC: 0.8874

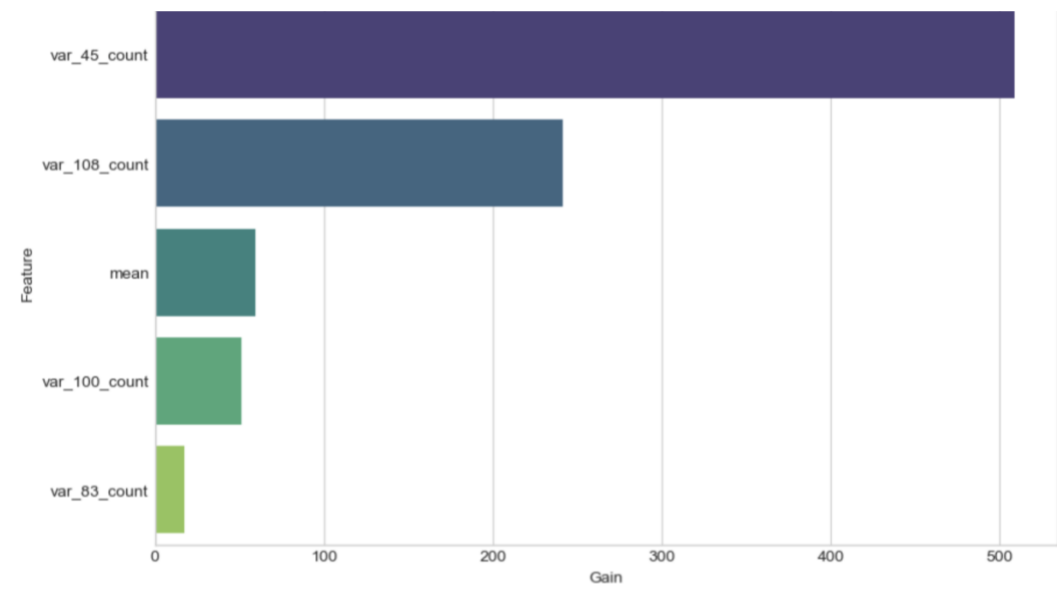


Figure 5. Gain-based feature ranking showing strong predictive contribution of engineered variables

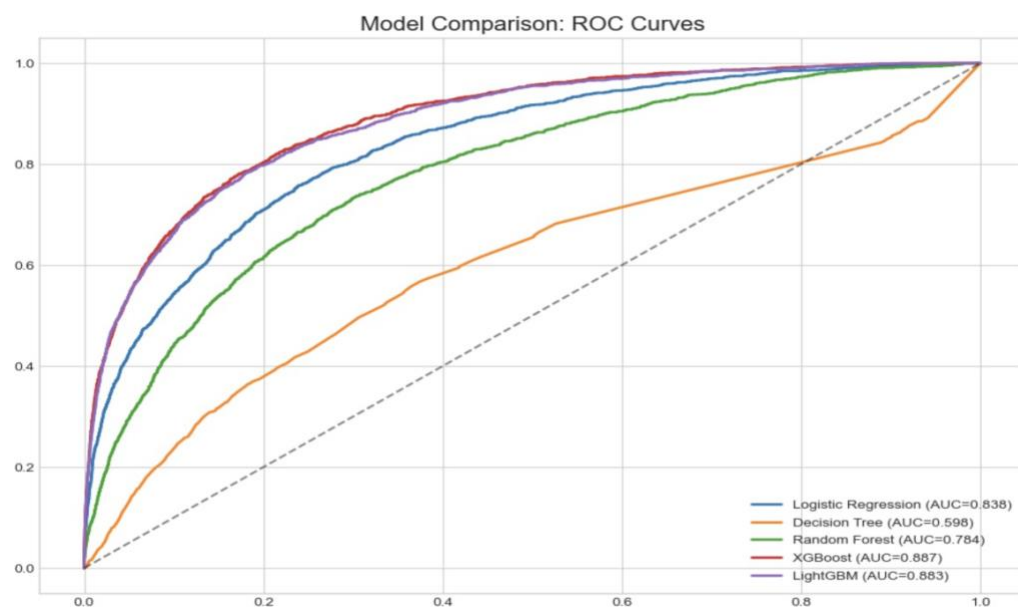


Figure 6. ROC comparison of Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM

Hypothesis Model 3: Ensemble Strategy (Bagging)

A Bagging (Bootstrap Aggregating) strategy was implemented to test if variance reduction could improve stability. This involved training multiple independent XGBoost models on different random subsets of the training data (undersampled bags) and averaging their predictions to produce a final probability score.

Resulting AUC: 0.8712

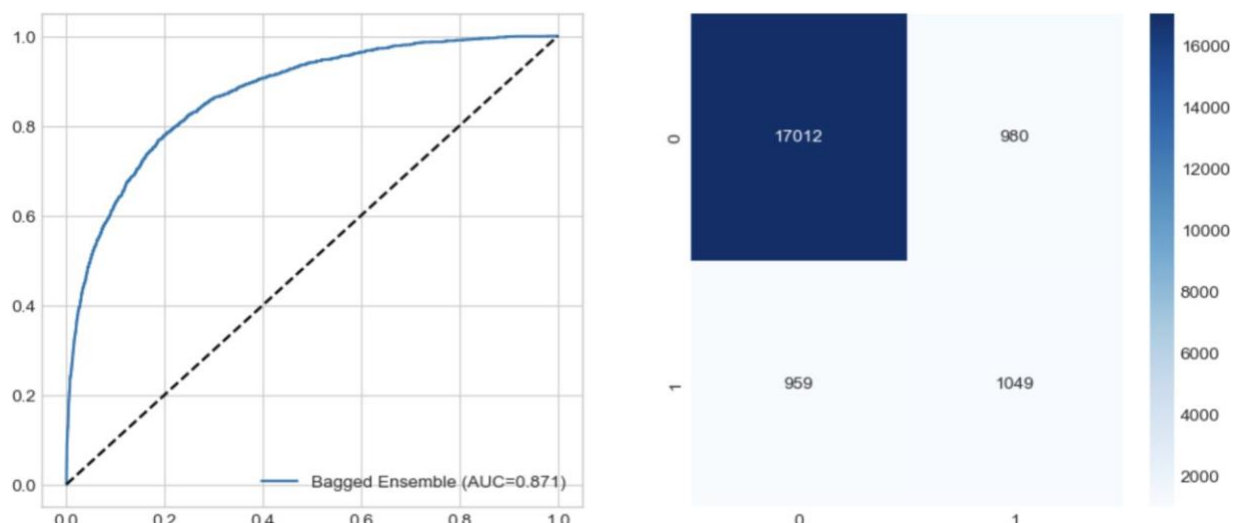


Figure 7. Bagging ensemble performance showing moderate improvement but rise in computation cost

Champion Model: Undersampled LightGBM with Gain-Based Feature Pruning

The final architecture transitioned to a Light Gradient Boosting Machine (LightGBM) to exploit its efficient leaf-wise tree growth strategy. Unlike previous iterations, this model removed `scale_pos_weight` and instead utilised Random Undersampling to create a strict 50:50 training distribution. This physical balancing prevented "double correction" bias. The pipeline was further optimised by applying Gain-Based Feature Pruning, which removed low-value features below 0.05 feature gain to reduce dimensionality to ~150 variables before final hyperparameter tuning.

Resulting AUC: 0.8826

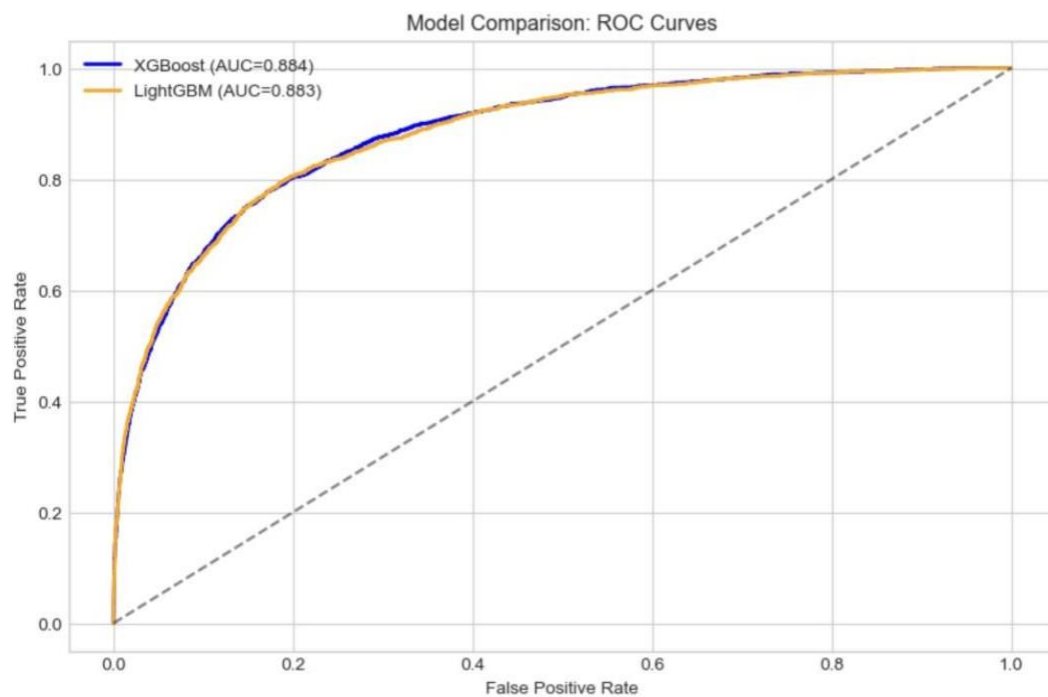


Figure 8. ROC curve showing LightGBM and XGBoost AUC

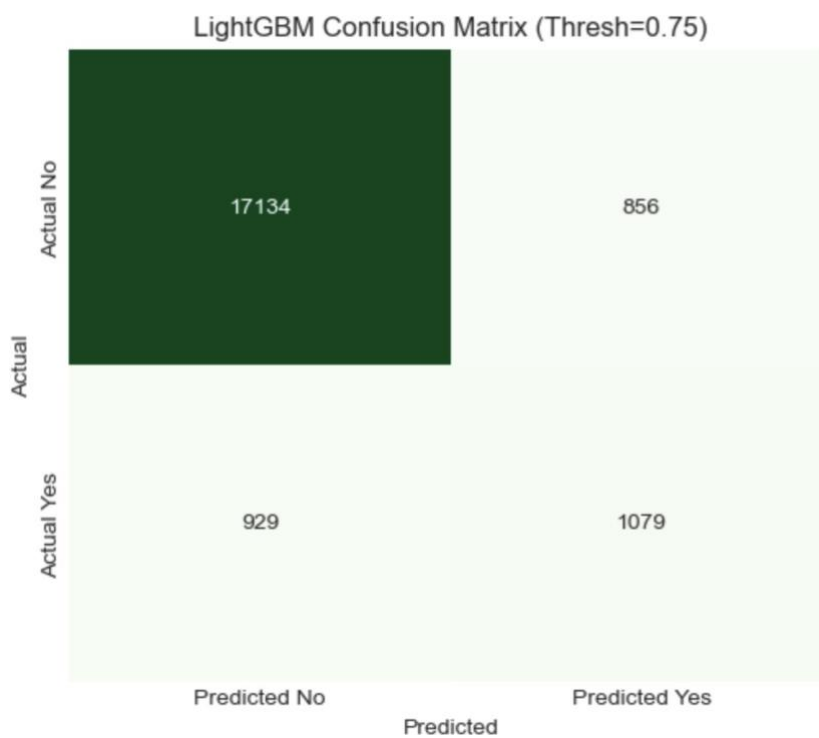


Figure 9. Confusion matrix at threshold 0.75 for LightGBM

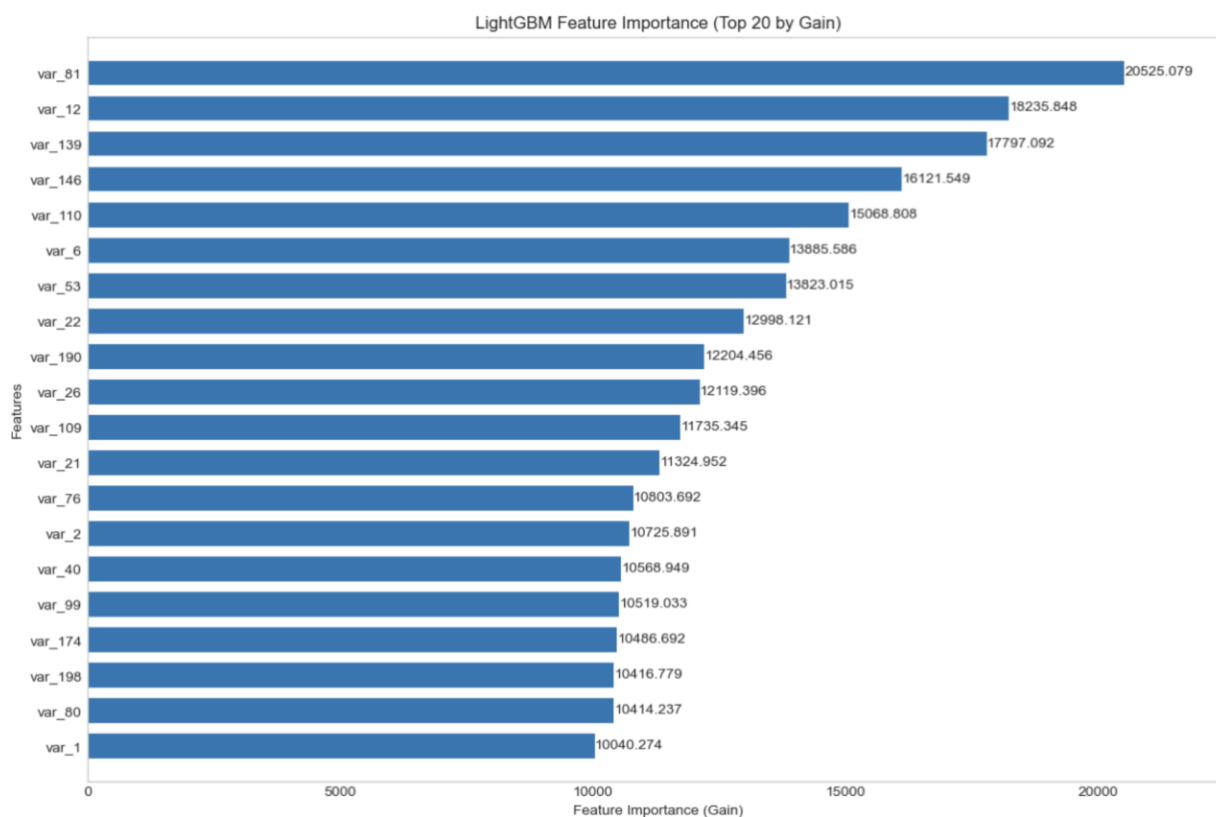


Figure 10. Top 20 features contributing to LightGBM predictions

This model balances interpretability, computational efficiency, and predictive power. LightGBM employs a leaf-wise tree growth strategy that captures complex nonlinear interactions more efficiently than level-wise growers, providing a key advantage when modelling anonymised numerical data.

4. Results and Evaluation

4.1 Comparative Model Performance and Model Rejection

This section presents a comparative evaluation of all models developed during experimentation, with performance assessed on the stratified 20% test set. Since all models naturally perform well on the dominant Class 0, overall accuracy is a poor metric. Therefore, model selection was driven strictly by performance on Class 1, focusing on the ability to capture actual applicants (Recall) without generating excessive false leads (Precision).

Table 1. Comparison of the results from various models

Model Architecture	Strategy	AUC Score	F1 Score	Precision	Recall	Operational Verdict
Logistic Regression	Baseline + Cost-Sensitive Learning	0.8595	0.49	0.46	0.52	Rejected: Failed to capture non-linear patterns; high false negative rate.
XGBoost	Feature Engineering + Cost-Sensitive	0.8874	0.54	0.54	0.55	Rejected: Highest raw predictive power, but high computational cost and lower interpretability hindered deployment.
Bagging	Bootstrap Aggregation (Undersampled)	0.8712	0.52	0.52	0.52	Rejected: Offered stability but yielded diminishing returns compared to the complexity of maintaining multiple models.
LightGBM (Champion)	Undersampling + Gain-Based Pruning	0.8826	0.55	0.56	0.54	Selected: Optimal balance of high precision (5.6x lift), fast inference

						speed, and explainability.
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4.2 Results Analysis

The baseline Logistic Regression model established the lower bound for performance. Using cost-sensitive learning, it achieved an AUC of 0.8595 and an F1 score of 0.49. While moderately effective in identifying minority-class customers (Recall = 0.52), its linear decision boundary failed to capture the rich, nonlinear relationships present in the anonymised features. This resulted in a high false-negative rate, making the model unsuitable as the primary solution.

Feature-engineered boosted models demonstrated the strongest individual predictive capability. XGBoost, enriched with row-based statistics and frequency-encoded columns, achieved the highest AUC (0.8874) and an F1 score of 0.54. This showed that incorporating statistical transformations helped uncover latent behavioural patterns. However, the approach required significantly higher computational resources and resulted in lower interpretability—both of which were considered disadvantages for real-world deployment.

The bagging ensemble, consisting of multiple undersampled XGBoost models aggregated via probability averaging, achieved stable performance (AUC = 0.8712, F1 = 0.52). While it reduced model variance, the complexity of training and maintaining multiple models did not justify its marginal gains compared with simpler boosted-tree architectures.

4.3 Rationale for Selecting LightGBM

LightGBM was selected as the champion model because it effectively optimises the trade-off between predictive accuracy and operational efficiency. By leveraging a pruned feature set of approximately 150 variables, the model achieves rapid inference speeds and significantly lower computational costs than XGBoost.

From a business perspective, LightGBM delivers a strong ROI by maximising precision, thereby reducing the financial waste associated with unnecessary outreach to non-applicants. Crucially, it maintains high recall for the minority class, ensuring the bank captures the majority of business opportunities. Finally, the model offers superior interpretability, allowing the management board to clearly trace feature importance back to business rationale.

4.4 Feature Importance and Error Pattern Insights

Gain-based feature pruning reduced the feature space from 200 to approximately 150 high-priority variables. LightGBM's feature importance consistently highlighted a small subset of variables as key drivers of the model's predictions. Although anonymised, these features likely represent core behavioural indicators (e.g., transaction intensity, variability, or aggregated activity patterns).

Error analysis revealed that most false negatives occurred among customers exhibiting weaker or noisier behavioural signals. False positives, on the other hand, often involved customers whose behaviour resembled applicants but who had not yet applied—suggesting that some misclassifications may represent valuable future leads rather than model failures.

5. Conclusion

This project aimed to develop a robust predictive model capable of identifying Universal Plus customers likely to submit business credit applications in the future. Working with a high-dimensional, heavily imbalanced dataset required careful preprocessing, feature engineering, and model comparison aligned with insights from the academic literature.

A comprehensive modelling pipeline was executed, progressing from baseline Logistic Regression through feature-engineered boosted trees and ensemble bagging. Logistic Regression delivered interpretability but failed to sufficiently capture complex behavioural patterns. Feature-engineered XGBoost achieved the highest individual AUC but introduced increased training time and reduced explainability.

The final champion model, an undersampled LightGBM with gain-based feature pruning, delivered the optimal balance of accuracy, efficiency, and interpretability. Achieving an AUC of 0.8826 and an F1 score of 0.55, it provides a substantial improvement over the baseline performance. At the optimised threshold of ~ 0.75 , the model achieves a precision of 56%, representing a 5.6 $\times$ lift compared with the base application rate. This ensures that more than half of contacted customers are genuinely likely to apply for credit, improving outreach efficiency and supporting informed relationship management. The model is now positioned for integration into the Universal Plus CRM system, where it will generate weekly propensity scores to guide Relationship Manager priorities.

Despite strong performance, limitations remain. Anonymised features restricted domain-driven feature engineering, and class imbalance continues to constrain precision ceilings. Future development should incorporate temporal behaviour, customer demographics, and interaction histories to refine the predictive boundary further. Overall, the project demonstrates that a carefully tuned ensemble machine learning approach provides a scalable and strategically valuable tool, enabling Universal Plus to transition toward proactive, data-driven customer engagement.

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